MTH316 2023S1 Exam

Feel free to contact me at maths_and_programming@outlook.sg if you need tuition. I provide university level tuition for the following SUSS units:

* ENG201 Linear Systems Analysis and Design
* ENG301 Microprocessor Programming
* ICT101 Computational Thinking
* ICT133 Structured Programming
* ICT162 Object Oriented Programming
* MTD215 Application of C++ in Multimedia
* ICT231 Data Structures and Algorithms I
* ICT232 Data Structures and Algorithms II
* MTH105 Fundamentals of Mathematics
* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH319 Mathematical Methods I
* MTH320 Mathematical Methods II
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference
* MTH355 Basic Mathematical Optimisation

Donaldson Tan

SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH316

Examination - January Semester 2023

Multivariable Calculus

Friday, 19 May 2023 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **SIX (6)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.
4. This is **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

MTH316 Copyright © 2023 Singapore University of Social Sciences (SUSS)

Answer all questions. (Total 100 marks)

Question 1

Define a function $f : \{(x, y) \in \mathbb{R}^2 \mid x \geq -1 \text{ and } (x, y) \neq (-1, 2)\} \longrightarrow \mathbb{R}$ by

$$f(x, y) = \frac{x + y^2 - 4y + 5}{\sqrt{x+1} + (y-2)^2}.$$

Question 1a(i)

Determine $\lim\limits_{(x,y)\to(1,-2)} f(x, y)$, if it exists. Otherwise, show that it does not exist.

(3 marks)

Question 1a(ii)

Determine $\lim\limits_{(x,y)\to(-1^+,2)} f(x, y)$, if it exists. Otherwise, show that it does not exist.

(8 marks)

Question 1b

Define a function $g : \{(x, y) \in \mathbb{R}^2 \mid x \geq \pi \text{ and } y \leq \pi\} \longrightarrow \mathbb{R}$ by

$$g(x, y) = \begin{cases} \dfrac{k(2\pi - x - y)}{\sqrt{x-\pi} + \sqrt{\pi - y}}, & (x, y) \neq (\pi, \pi), \\ 0, & (x, y) = (\pi, \pi). \end{cases}$$

Determine the value(s) of k for the function g to be continuous at (π, π).

(9 marks)

Question 2

A solid region E is described by

$$E : \left\{(x, y, z) | -9 \leq z \leq 3y,\ x^2 + y^2 \leq 9\right\}.$$

Question 2a

Sketch the projections of region E onto the (x, y) and (y, z) coordinate planes.

(6 marks)

Question 2b

Set up an integral in Cartesian coordinates that gives the volume of region E. Do not evaluate the integral.

(4 marks)

Question 2c

Convert the integral in Question 2b to one in cylindrical coordinates and evaluate it to compute the volume of solid E.

(10 marks)

Question 3

In a laboratory where testing equipment is very sensitive to moisture, the relative humidity (RH) of the air is carefully controlled. The RH (%) in the laboratory on a test bench can be modelled by the function

$$f(x, y) = x^3 - \frac{x^2}{2} + \frac{y^2}{4} + 32,$$

where the distances x and y (measured in m) from one corner of the test bench satisfy the inequalities $0 \leq x \leq 3,\ 0 \leq y \leq 6$.

Question 3a

Calculate the gradient of the RH function at the position (1, 2). Hence or otherwise, find a unit vector in the direction of maximum RH decrease at the position (1, 2).

(5 marks)

Question 3b

A robotic arm moves a test specimen from the position (1, 2) to a new position (2, 2). If the path of the motion is a straight line, what is the gradient of RH experienced by the specimen along its line of motion at its initial position?

(5 marks)

Question 3c

Suppose now the robotic arm moves along the path $y = 2x$. Apply Lagrange multipliers to find the location of the lowest RH along this new path on the test bench.

(10 marks)

Question 4

A force field is defined by

$$\mathbf{F}(x, y) = \begin{bmatrix} 2x(y-1) \\ 3x^2 + y \end{bmatrix}.$$

$\frac{\partial Q}{\partial x} = 6x$

$\frac{\partial P}{\partial y} = 2x$

Three paths C_1, C_2 and C_3 that forms an enclosed region are shown below.

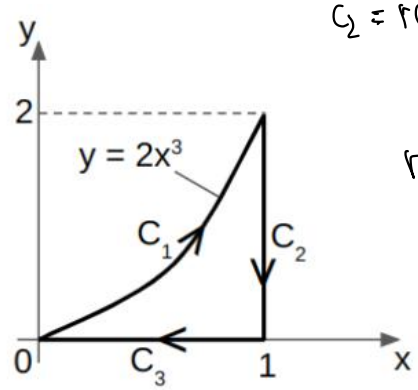


Figure Q4

$C_2 = r(t) = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + t\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} 1 \\ 2+t \end{bmatrix}$

$r'(t) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$F(t) = [$

$r(t) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -1 \\ 0 \end{bmatrix}$

$= \begin{bmatrix} 1-t \\ 0 \end{bmatrix}$

Question 4a

Compute directly the work done by $F(x, y)$ along the path C_1.

(7 marks)

Question 4b

Use Green's theorem to evaluate the work done by $F(x, y)$ along the combined path $C_2 + C_3$.

(13 marks)

Question 5

Define a vector field F by

$$\mathbf{F}(x, y, z) = \begin{bmatrix} 3x \\ y + \sin z \\ z/2 \end{bmatrix}.$$

Question 5a

Let S_1 be a circular disc of radius 3 that lies on the plane $z = 0$, where the center of the disc is at (0, 0, 0). By orientating S_1 with a downward normal vector, compute the flux of $F(x, y, z)$ through S_1.

(5 marks)

Question 5b

Define S_2 as another surface that lies on the plane $z - x = 3$ and is bounded by the cylinder $x^2 + y^2 = 9$. By orientating S_2 with a normal vector having an upward component, evaluate the flux of $F(x, y, z)$ through S_2.

(10 marks)

Question 5c

Define another surface S_3 as the portion of the cylindrical surface $x^2 + y^2 = 9$ that is bounded by the plane $z = 0$ from below and the plane $z - x = 3$ from the top. Orientate S_3 with outward-pointing normal vectors. Use the divergence theorem to evaluate the flux of $F(x, y, z)$ through S_3.

(5 marks)

----- END OF PAPER -----

1. (a) i. $f(x,y) = \dfrac{x+y^2-4y+5}{\sqrt{x+1}+(y-2)^2}$

$$\lim_{(x,y)\to(1,-2)} f(x,y) = \lim_{(x,y)\to(1,-2)} \frac{x+y^2-4y+5}{\sqrt{x+1}+(y-2)^2}$$

$$= \frac{1+4-4(-2)+5}{\sqrt{1+1}+(-4)^2}$$

$$= \frac{18}{\sqrt{2}+16}$$

(ii) moving along $y=2$:

$$\lim_{(x,y)\to(-1^+,2)} f(x,y) = \lim_{x\to-1^+} f(x,2)$$

$$= \lim_{x\to-1^+} \frac{x+4-8+5}{\sqrt{x+1}+0^2}$$

$$= \lim_{x\to-1^+} \frac{x+1}{\sqrt{x+1}}$$

$$= \lim_{x\to-1^+} \sqrt{x+1}$$

$$= 0$$

moving along $x=-1$:

$$\lim_{(x,y)\to(-1^+,2)} f(x,y) = \lim_{y\to2} f(-1,y)$$

$$= \lim_{y\to2} \frac{-1+y^2-4y+5}{0+(y-2)^2}$$

$$= \lim_{y\to2} \frac{y^2-4y+4}{(y-2)^2}$$

$$= \lim_{y\to2} \frac{(y-2)^2}{(y-2)^2}$$

$$= 1$$

$\therefore$ Hence, the limit doesn't exist

1(b) for $g(x,y)$ to be continuous,

$$\lim_{(x,y)\to(\pi,\pi)} g(x,y) = g(\pi,\pi) = 0$$

$$\lim_{(x,y)\to(\pi,\pi)} g(x,y) = \lim_{(x,y)\to(\pi,\pi)} \frac{k(2\pi-x-y)}{\sqrt{x-\pi}+\sqrt{\pi-y}}$$

$$= \lim_{(x,y)\to(\pi,\pi)} \frac{k(2\pi-x-y)}{\sqrt{x-\pi}+\sqrt{\pi-y}} \cdot \frac{\sqrt{x-\pi}-\sqrt{\pi-y}}{\sqrt{x-\pi}-\sqrt{\pi-y}}$$

$$= \lim_{(x,y)\to(\pi,\pi)} \frac{k(2\pi-x-y)(\sqrt{x-\pi}-\sqrt{\pi-y})}{(x-\pi)-(\pi-y)}$$

$$= \lim_{(x,y)\to(\pi,\pi)} \frac{-k(x+y-2\pi)(\sqrt{x-\pi}-\sqrt{\pi-y})}{x+y-2\pi}$$

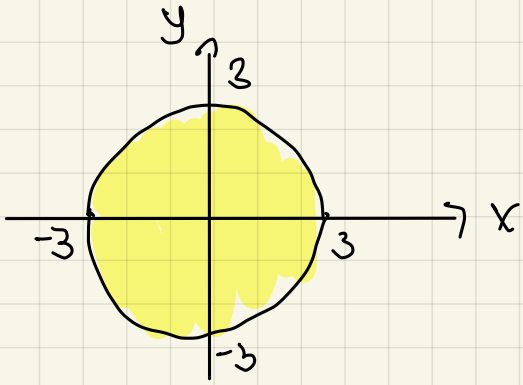
$$= \lim_{(x,y)\to(\pi,\pi)} -k\left[\sqrt{x-\pi}-\sqrt{\pi-y}\right]$$

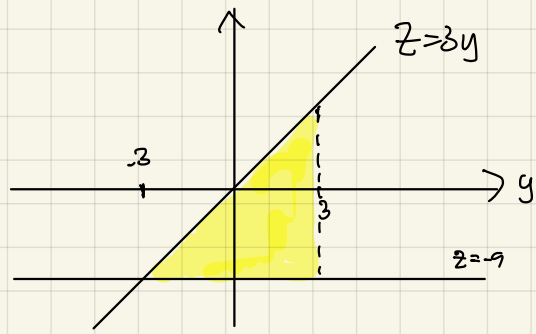
$$= -k\left(\sqrt{\pi-\pi}-\sqrt{\pi-\pi}\right)$$

$$= 0$$

$\therefore$ k is any real value

2(a) $E = \{(x,y,z) : -9 \le z \le 3y,\ x^2+y^2 \le 9\}$





(b) volume $= \iiint_E 1\, dV$

$$= \int_{-3}^{3}\int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}}\int_{-9}^{3y} 1\, dz\, dy\, dx$$

(c) $x = r\cos\theta,\quad y = r\sin\theta$

$dx\,dy = r\,dr\,d\theta$

$$\text{volume} = \iiint_E 1\, dV$$

$$= \int_0^3\int_0^{2\pi}\int_{-9}^{3r\sin\theta} r\ dz\, d\theta\, dr$$

$$= \int_0^3\int_0^{2\pi} r(3r\sin\theta + 9)\, d\theta\, dr$$

$$= \int_0^3\int_0^{2\pi} 3r^2\sin\theta + 9r\ d\theta\, dr$$

$$= \int_0^3 \left[9r\theta - 3r^2\cos\theta\right]_{\theta=0}^{\theta=2\pi} dr$$

$$= \int_0^3 (18\pi r - 3r^2) - (0 - 3r^2)\ dr$$

$$= \int_0^3 18\pi r\ dr$$

$$= \left[9\pi r^2\right]_0^3$$

$$= 9\pi(9-0)$$

$$= 81\pi$$

3(a) $f(x,y) = x^3 - \frac{x^2}{2} + \frac{y^2}{4} + 32$ for $x \in [0,3]$, $y \in [0,6]$

$\frac{\partial f}{\partial x} = 3x^2 - x \qquad \frac{\partial f}{\partial y} = \frac{y}{2}$.

$\nabla f(x,y) = \begin{bmatrix} 3x^2 - x \\ \frac{y}{2} \end{bmatrix} \qquad \nabla f(1,2) = \begin{bmatrix} 3-1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$D_f u(x,y) = |\nabla f(x,y)|\,|u| \cos\theta$

$\Rightarrow \min D_f u(x,y) = -|\nabla f(x,y)|$

$\Rightarrow \underset{\sim}{u}$ must be a unit vector in the opposite direction of $\nabla f(1,2)$

$|\nabla f(1,2)| = \sqrt{2^2 + (1)^2} = \sqrt{5}$

$\therefore \underset{\sim}{u} = -\frac{1}{|\nabla f(1,2)|} \nabla f(1,2) = -\frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

(b) $\underset{\sim}{u} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\nabla f(1,2) = \frac{2}{\sqrt{5}} \begin{bmatrix} 2 \\ \frac{1}{2} \end{bmatrix}$

rate of change $= D_u f(1,2)$

$= \nabla f(1,2) \cdot \underset{\sim}{u}$

$= \begin{bmatrix} 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$= 2$

3(c) Minimise $f(x,y)$

subject to $g(x,y) = 0$ where $g(x,y) = y - 2x$

$\frac{\partial g}{\partial x} = -2, \quad \frac{\partial g}{\partial y} = 1$

let $\nabla f = \lambda \nabla g$.

$\begin{bmatrix} 3x^2 - x \\ \frac{y}{2} \end{bmatrix} = \lambda \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

$\Rightarrow \lambda = \frac{3x^2 - x}{-2} = \frac{y}{2}$

$3x^2 - x = -y$

$y = x - 3x^2$

$y - 2x = -x - 3x^2$

$0 = -x - 3x^2$

$0 = 3x^2 + x = x(3x+1)$

$x = 0$ or $-\frac{1}{3}$ [reject]

$y = 2x = 0$

$f(0,0) = 32$

The location of the lowest pit is $(0,0)$

4(a) C_1: $r(t) = \begin{bmatrix} t \\ 2t^3 \end{bmatrix}$ $\quad 0 \le t \le 1 \quad$ $r'(t) = \begin{bmatrix} 1 \\ 6t^2 \end{bmatrix}$

$$F(t) = F(t, 2t^3) = \begin{bmatrix} 2t(2t^3-1) \\ 3t^2 + 2t^3 \end{bmatrix}$$

$$\begin{aligned}
\text{work done} &= \int_{C_1} F \cdot dr \\
&= \int_0^1 F(t) \cdot r'(t)\, dt \\
&= \int_0^1 \begin{bmatrix} 2t(2t^3-1) \\ 3t^2+2t^3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 6t^2 \end{bmatrix} dt \\
&= \int_0^1 2t(2t^3-1) + 6t^2(3t^2+2t^3)\, dt \\
&= \int_0^1 4t^4 - 2t + 18t^4 + 12t^5\, dt \\
&= \int_0^1 12t^5 + 22t^4 - 2t\, dt \\
&= \left[2t^6 + \frac{22}{5} t^5 - t^2 \right]_0^1 \\
&= 2 + \frac{22}{5} - 1 \\
&= \frac{27}{5}
\end{aligned}$$

4(b) $F(x,y) = \begin{bmatrix} 2x(y-1) \\ 3x^2+y \end{bmatrix}$

$\frac{\partial Q}{\partial x} = 6x \qquad \frac{\partial P}{\partial y} = 2x$

By Green's Theorem

$$\begin{aligned}
\int_{C_1+C_2+C_3} F \cdot dr &= -\iint_S \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\, dA \\
&= -\iint_S 6x - 2x\, dA \\
&= -\iint_S 4x\, dA \\
&= -\int_0^1 \int_0^{2x^3} 4x\, dy\, dx \\
&= -\int_0^1 \int_0^{2x^3} 4x\, dy\, dx \\
&= -\int_0^1 4x[y]_0^{2x^3}\, dx \\
&= -\int_0^1 8x^4\, dx \\
&= -\left[\frac{8}{5} x^5 \right]_0^1 \\
&= -\frac{8}{5}
\end{aligned}$$

the path is clockwise

$$\begin{aligned}
\int_{C_1+C_2} F \cdot dr &= \int_{C_1+C_2+C_3} F \cdot dr - \int_{C_1} F \cdot dr \\
&= -\frac{8}{5} - \frac{27}{5} \\
&= -\frac{35}{5} \\
&= -7
\end{aligned}$$

$\Rightarrow$

5. $F(x,y,z) = \begin{bmatrix} 3x \\ y + \sin z \\ z/2 \end{bmatrix}$

(a) S_1: $r(\rho,\theta) = \begin{bmatrix} \rho\cos\theta \\ \rho\sin\theta \\ 0 \end{bmatrix}$ where $0 \le \rho \le 3$, $0 \le \theta \le 2\pi$ $\qquad F(\rho,\theta) = \begin{bmatrix} 3\rho\cos\theta \\ \rho\sin\theta \\ 0 \end{bmatrix}$

$$r_\rho = \begin{bmatrix} \cos\theta \\ \sin\theta \\ 0 \end{bmatrix} \qquad r_\theta = \begin{bmatrix} -\rho\sin\theta \\ \rho\cos\theta \\ 0 \end{bmatrix}$$

$$r_\rho \times r_\theta = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ \cos\theta & \sin\theta & 0 \\ -\rho\sin\theta & \rho\cos\theta & 0 \end{vmatrix} = \begin{bmatrix} 0 \\ 0 \\ \rho\cos^2\theta + \rho\sin^2\theta \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \rho \end{bmatrix}$$

flux of F through $S_1 = -\iint_{-S_1} F \cdot (r_\rho \times r_\theta)\, d\rho\, d\theta$

$$= -\int_0^3 \int_0^{2\pi} \begin{bmatrix} 3\rho\cos\theta \\ \rho\sin\theta \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ \rho \end{bmatrix} d\theta\, d\rho$$

$$= -\int_0^3 \int_0^{2\pi} 0 \; d\theta\, d\rho$$

$$= 0$$

(b) S_2: $r(\rho,\theta) = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ x+3 \end{bmatrix} = \begin{bmatrix} \rho\cos\theta \\ \rho\sin\theta \\ \rho\cos\theta + 3 \end{bmatrix}$ for $0 \le \rho \le 3$, $0 \le \theta \le 2\pi$

$$r_\rho = \begin{bmatrix} \cos\theta \\ \sin\theta \\ \cos\theta \end{bmatrix} \quad r_\theta = \begin{bmatrix} -\rho\sin\theta \\ \rho\cos\theta \\ -\rho\sin\theta \end{bmatrix} \quad r_\rho \times r_\theta = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ \cos\theta & \sin\theta & \cos\theta \\ -\rho\sin\theta & \rho\cos\theta & -\rho\sin\theta \end{vmatrix} = \begin{bmatrix} -\rho \\ 0 \\ \rho \end{bmatrix}$$

$$F(\rho,\theta) = \begin{bmatrix} 3\rho\cos\theta \\ \rho\sin\theta + \sin(\rho\cos\theta + 3) \\ \frac{1}{3}\rho\cos\theta + \frac{3}{2} \end{bmatrix}$$

flux through $S_2 = \iint_{S_2} F \cdot (r_\rho \times r_\theta)\, d\rho\, d\theta$

$$= \int_0^{2\pi} \int_0^3 -3\rho^2\cos\theta + \frac{\rho^2}{2}\cos\theta + \frac{3}{2}\rho \; d\rho\, d\theta$$

$$= \int_0^{2\pi} \left[-\rho^3\cos\theta + \frac{\rho^3}{6}\cos\theta + \frac{3}{4}\rho^2 \right]_{\rho=0}^{\rho=3} d\theta$$

$$= \int_0^{2\pi} \frac{27}{4} - \frac{45}{2}\cos\theta \; d\theta$$

$$= \left[\frac{27}{4}\theta - \frac{45}{2}\sin\theta \right]_0^{2\pi}$$

$$= \frac{27}{2}\pi$$

5(c) $\operatorname{div} F = \nabla \cdot F$

$$= \frac{\partial}{\partial x}(3x) + \frac{\partial}{\partial y}(y + \sin z) + \frac{\partial}{\partial z}\left(\frac{z}{2}\right)$$

$$= 3 + 1 + \frac{1}{2}$$

$$= \frac{9}{2}$$

consider the limits of integration

$$0 \leq z \leq 3 + x$$

$$\Rightarrow 0 \leq z \leq 3 + r\cos\theta$$

$$0 \leq r \leq 3$$

$$0 \leq \theta \leq 2\pi$$

$$\iint_{S_3} F \cdot dS = \iiint_E \operatorname{div} F \, dV$$

$$= \iiint_E \frac{9}{2} \, dV$$

$$= \int_0^{2\pi} \int_0^3 \int_0^{3+r\cos\theta} \frac{9}{2} r \, dz \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^3 \frac{9}{2} r \left[z \right]_0^{3+r\cos\theta} dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^3 \frac{27}{2} r + \frac{9}{2} r^2 \cos\theta \, dr \, d\theta$$

$$= \int_0^{2\pi} \left[\frac{27}{4} r^2 + \frac{3}{2} r^3 \cos\theta \right]_{r=0}^{r=3} d\theta$$

$$= \int_0^{2\pi} 81\cos\theta + \frac{243}{4} \, d\theta$$

$$= \left[\frac{243}{4}\theta + \frac{81}{2}\sin\theta \right]_0^{2\pi}$$

$$= \frac{243}{2}\pi$$

flux through $S_3 = \iiint_E \operatorname{div} F \, dV - \iint_{S_1} F \cdot dS - \iint_{S_2} F \cdot dS$

$$= \frac{243}{2}\pi - 0 - \frac{27}{2}\pi$$

$$= 118\pi$$